The normal equations give:

$$a = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{987}{5} = 197.4$$

$$b = \frac{\sum_{i=1}^{n} u_i y_i}{\sum_{i=1}^{n} u_i^2} = \frac{162}{10} = 16.2$$

$y = 197.4 + 16.2\ (X - 2)$

$= 197.4 + 16.2\ X - 32.4$

$= 16.2\ X + 165$

That is, $y = 165 + 16.2X$

To get the required trend values, put $X = 0, 1, 2, 3, 4$ in the estimated equation.

$X = 0,\ y = 165 + 0 = 165$

$X = 1,\ y = 165 + 16.2 = 181.2$

$X = 2,\ y = 165 + 32.4 = 197.4$

$X = 3,\ y = 165 + 48.6 = 213.6$

$X = 4,\ y = 165 + 64.8 = 229.8$

Hence, the trend values for 2010, 2011, 2012, 2013 and 2014 are 165, 181.2, 197.4, 213.6 and 229.8 respectively.

Steps for calculating trend values when *n* is even:

i). Subtract the first year from all the years (x)

ii). Find $u_i = 2X - (n - 1)$

iii). Find u_i^2 and $u_i y_i$

Then follow the same procedure used in previous method for odd years

Example 7.8

Tourist arrivals (Foreigners) in Tamil Nadu for 6 consecutive years are given in the following table. Calculate the trend values by using the method of least squares.

Year	2005	2006	2007	2008	2009	2010
No. of arrivals (in lakhs)	12	13	18	20	24	28

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